

11.52 Give a 95% prediction interval for the prediction you selected in the preceding exercise.

11.5 Examining Lack of Fit in Linear Regression

In our study of linear regression, we have been concerned with how well a linear regression model $y = \beta_0 + \beta_1 x + \varepsilon$ fits, but only from an intuitive standpoint. We could examine a scatterplot of the data to see whether it looked linear and we could test whether the slope differed from 0; however, we had no way of testing to see whether a higher-order model would be a more appropriate model for the relationship between y and x . This section will outline situations in which we can test for the validity of a linear regression model.

Pictures (or graphs) are always a good starting point for examining lack of fit. First, use a scatterplot of y versus x . Second, a plot of residuals $y_i - \hat{y}_i$ versus predicted values $\hat{y}_i$ may give an indication of the following problems:

1. Outliers or erroneous observations. In examining the residual plot, your eye will naturally be drawn to data points with unusually high (in absolute value) residuals.
2. Violation of the assumptions. For the model $y = \beta_0 + \beta_1 x + \varepsilon$, we have assumed a linear relation between y and the dependent variable x , and independent, normally distributed errors with a constant variance.

The residual plot for a model and data set that has none of these apparent problems would look much like the plot in Figure 11.14. Note from this plot that there are no extremely large residuals (and hence no apparent outliers) and there is no trend in the residuals to indicate that the linear model is inappropriate. When a higher-order model is more appropriate, a residual plot more like that shown in Figure 11.15 would be observed.

FIGURE 11.14
Residual plot with no
apparent pattern

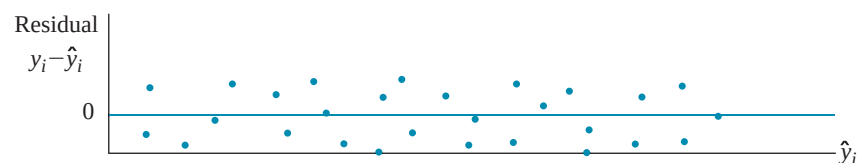
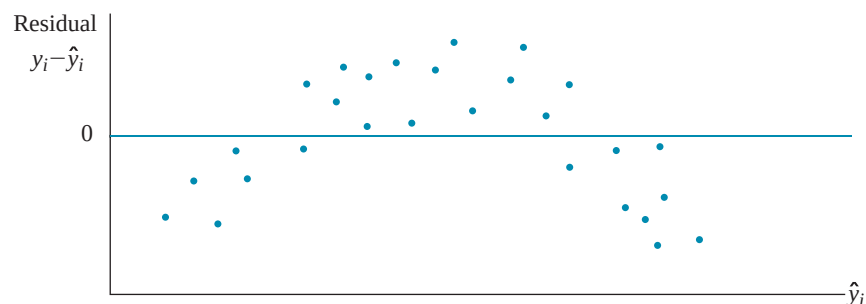


FIGURE 11.15
Residual plot showing the
need for a higher-order
model



A check of the constant variance assumption can be addressed in the y versus x scatterplot or with a plot of the residuals ($y_i - \hat{y}_i$) versus x_i . For example, a pattern of residuals as shown in Figure 11.16 indicates homogeneous error variances across values of x ; Figure 11.17 indicates that the error variances increase with increasing values of x .

FIGURE 11.16
Residual plot showing
homogeneous error variances

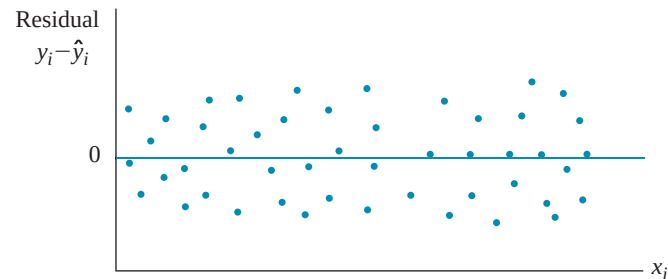
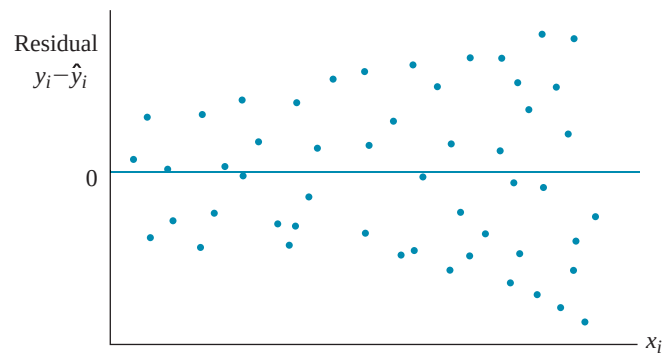


FIGURE 11.17
Residual plot showing error
variances increasing with x



The question of independence of the errors and normality of the errors is addressed later in Chapter 13. We illustrate some of the points we have learned so far about residuals by way of an example.

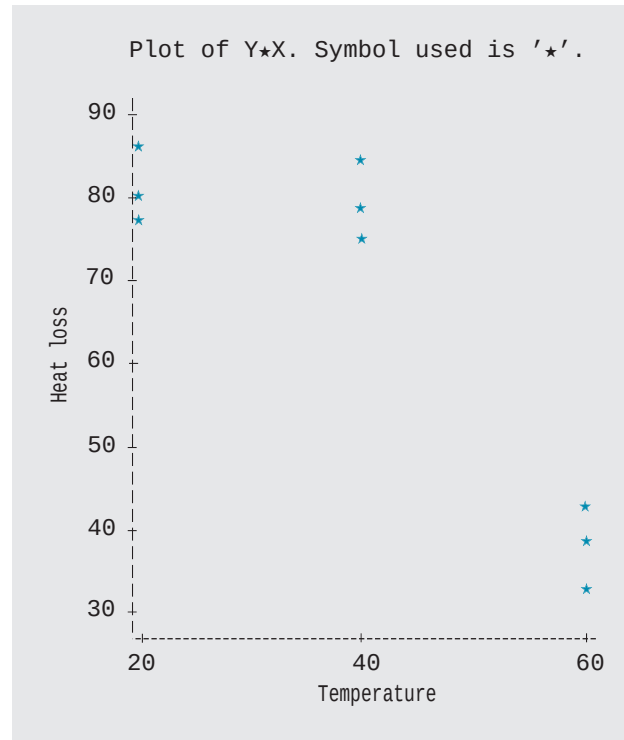
EXAMPLE 11.11

The manufacturer of a new brand of thermal panes examined the amount of heat loss by random assignment of three different panes to each of the three outdoor temperature settings being considered. For each trial, the window temperature was controlled at 68°F and 50% relative humidity.

Outdoor Temperature (°F)	Heat Loss
20	86, 80, 77
40	78, 84, 75
60	33, 38, 43

- Plot the data.
- Fit the linear regression model $y = \beta_0 + \beta_1 x + \varepsilon$ and test $H_0: \beta_1 = 0$ (give the p -value for your test).
- Compute $\hat{y}_i$ and $y_i - \hat{y}_i$ for the nine observations. Plot $y_i - \hat{y}_i$ versus $\hat{y}_i$.
- Does the constant variance assumption seem reasonable?

Solution The computer output shown here can be used to address the four parts of this example.



Dependent Variable: Y HEAT LOSS

Analysis of Variance

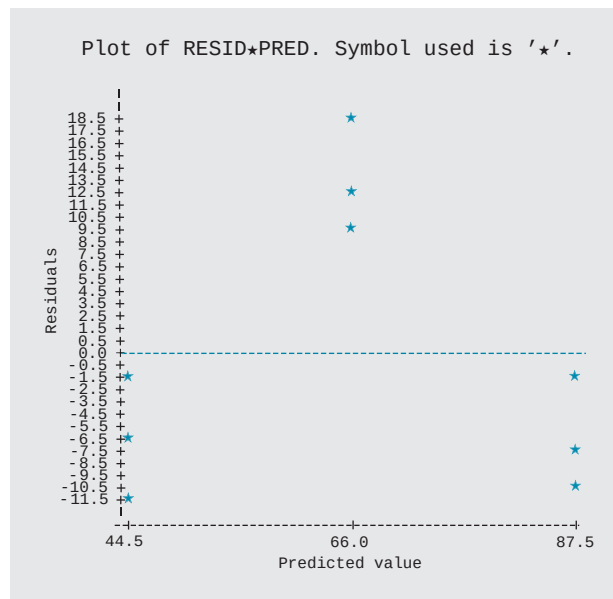
Source	DF	Sum of Squares	Mean Square	F Value	Prob>F
Model	1	2773.50000	2773.50000	21.704	0.0023
Error	7	894.50000	127.78571		
C Total	8	3668.00000			

Root SE	11.30423	R-square	0.7561
Dep Mean	66.00000	Adj R-sq	0.7213
C.V.	17.12763		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	T for H0: Parameter=0	Prob > T
INTERCEP	1	109.000000	9.96939762	10.933	0.0001
X	1	-1.075000	0.23074672	-4.659	0.0023

OBS	X	Y	PRED	RESID
1	20	86	87.5	-1.5
2	20	80	87.5	-7.5
3	20	77	87.5	-10.5
4	40	78	66.0	12.0
5	40	84	66.0	18.0
6	40	75	66.0	9.0
7	60	33	44.5	-11.5
8	60	38	44.5	-6.5
9	60	43	44.5	-1.5



- The scatterplot of y versus x certainly shows a downward linear trend, and there may be evidence of curvature as well.
- The linear regression model seems to fit the data well, and the test of $H_0: \beta_1 = 0$ is significant at the $p = .0023$ level. However, is this the best model for the data?
- The plot of residuals $(y_i - \hat{y}_i)$ against the predicted values $\hat{y}_i$ is similar to Figure 11.15, suggesting that we may need additional terms in our model.
- Because residuals associated with $x = 20$ (the first three), $x = 40$ (the second three), and $x = 60$ (the third three) are easily located, we really do not need a separate plot of residuals versus x to examine the constant variance assumption. It is clear from the original scatterplot and the residual plot shown that we do not have a problem.

How can we test for the apparent lack of fit of the linear regression model in Example 11.11? When there is more than one observation per level of the independent variable, we can conduct a test for lack of fit of the fitted model by partitioning SS (Residuals) into two parts, one **pure experimental error** and the other **lack of fit**. Let y_{ij} denote the response for the j th observation at the i th level

pure experimental error
lack of fit

of the independent variable. Then, if there are n_i observations at the i th level of the independent variable, the quantity

$$\sum_j (y_{ij} - \bar{y}_i)^2$$

provides a measure of what we will call pure experimental error. This sum of squares has $n_i - 1$ degrees of freedom.

Similarly, for each of the other levels of x , we can compute a sum of squares due to pure experimental error. The pooled sum of squares

$$SSP_{\text{exp}} = \sum_{ij} (y_{ij} - \bar{y}_i)^2$$

called the sum of squares for pure experimental error, has $\sum_i (n_i - 1)$ degrees of freedom. With SS_{Lack} representing the remaining portion of SSE, we have

$$SS(\text{Residuals}) = \underset{\text{error}}{\text{due to pure experimental}} \frac{SSP_{\text{exp}}}{\text{error}} + \underset{\text{to fit}}{\text{due to lack}} \frac{SS_{\text{Lack}}}{\text{to fit}}$$

If $SS(\text{Residuals})$ is based on $n - 2$ degrees of freedom in the linear regression model, then SS_{Lack} will have $df = n - 2 - \sum_i (n_i - 1)$.

Under the null hypothesis that our model is correct, we can form independent estimates of σ_{ϵ}^2 , the model error variance, by dividing SSP_{exp} and SS_{Lack} by their respective degrees of freedom; these estimates are called **mean squares** and are denoted by MSP_{exp} and MS_{Lack} , respectively.

The test for lack of fit is summarized here.

mean squares

A Test for Lack of Fit in Linear Regression

H_0 : A linear regression model is appropriate.

H_a : A linear regression model is not appropriate.

$$\text{T.S.: } F = \frac{MS_{\text{Lack}}}{MSP_{\text{exp}}},$$

where

$$MS_{\text{exp}} = \frac{SSP_{\text{exp}}}{\sum (n_i - 1)} = \frac{\sum_{ij} (y_{ij} - \bar{y}_i)^2}{\sum_i (n_i - 1)}$$

and

$$MS_{\text{Lack}} = \frac{SS(\text{Residuals}) - SSP_{\text{exp}}}{n - 2 - \sum (n_i - 1)}$$

R.R.: For specified value of α , reject H_0 (the adequacy of the model) if the computed value of F exceeds the table value for $df_1 = n - 2 - \sum_i (n_i - 1)$ and $df_2 = \sum_i (n_i - 1)$.

Conclusion: If the F test is significant, this indicates that the linear regression model is inadequate. A nonsignificant result indicates that there is insufficient evidence to suggest that the linear regression model is inappropriate.

EXAMPLE 11.12

Refer to the data of Example 11.11. Conduct a test for lack of fit of the linear regression model.

Solution It is easy to show that the contributions to experimental error for the differential levels of x are as shown here.

Level of x	$\bar{y}_i$	Contribution to Pure Experimental Error	
		$\sum_i (y_{ij} - \bar{y}_i)^2$	$n_i - 1$
20	81	42	2
40	79	42	2
60	38	50	2
Total		134	6

Summarizing these results, we have

$$SSP_{\text{exp}} = \sum_{ij} (y_{ij} - \bar{y}_i)^2 = 134$$

The output shown for Example 11.11 gives $SS(\text{Residual}) = 894.5$; hence, by subtraction,

$$SS_{\text{Lack}} = SS(\text{Residual}) - SSP_{\text{exp}} = 894.5 - 134 = 760.5$$

The sum of squares due to pure experimental error has $\sum_i (n_i - 1) = 6$ degrees of freedom; it therefore follows that with $n = 9$, SS_{Lack} has $n - 2 - \sum_i (n_i - 1) = 1$ degree of freedom. We find that

$$MSP_{\text{exp}} = \frac{SSP_{\text{exp}}}{6} = \frac{134}{6} = 22.33$$

and

$$MS_{\text{Lack}} = \frac{SS_{\text{Lack}}}{1} = 760.5$$

The F statistic for the test of lack of fit is

$$F = \frac{MS_{\text{Lack}}}{MSP_{\text{exp}}} = \frac{760.5}{22.33} = 34.06$$

Using $df_1 = 1$, $df_2 = 6$, and $\alpha = .05$, we will reject H_0 if $F \geq 5.99$.

Because the computed value of F exceeds 5.99, we reject H_0 and conclude that there is significant lack of fit for a linear regression model. The scatterplot shown in Example 11.11 confirms this nonlinearity.

To summarize: In situations for which there is more than one y -value at one or more levels of x , it is possible to conduct a formal test for lack of fit of the linear regression model. This test should precede any inferences made using the fitted linear regression line. If the test for lack of fit is significant, some higher-order polynomial in x may be more appropriate. A scatterplot of the data and a residual plot from the linear regression line should help in selecting the appropriate